SOM

- Distancia entre neuronas

$$d(P,W) = \sqrt{\sum_{i=1}^{M} (W_{ij} - P_i)^2}$$

- Actualización de pesos

$$W_{ij}(t+1) = W_{ij}(t) + \alpha h \left[P_i - W_{ij}(t)\right]$$

- Si la función de vecindad es nula solo se actualizan los pesos de la neurona ganadora, caso contrario, todos

Ejercicio 01.1 - Guía → Para todos los ejercicios del punto 1: $\alpha = 0,5$ $h = 1$

$$e_1 = (-0,54 ; 0,36) \qquad d(e_i^{(k)}, w_{ji}) = \sqrt{\sum_{i=1}^{M} (e_i^{(k)} - w_{ji})^2}$$

$$|d(P_1, W_1)| = \sqrt{(e_{11} - w_{11})^2 + (e_{12} - w_{12})^2} = 1,231$$

$$|d(e_1, W_2)| = \sqrt{(e_{12} - w_{21})^2 + (e_{22} - w_{22})^2} = 0,441 \leftarrow \text{GANADORA}$$

$$|d(e_1, W_3)| = \sqrt{(e_{11} - w_{31})^2 + (e_{12} - w_{32})^2} = 1,119$$

$$|d(e_1, W_4)| = \sqrt{(e_{11} - w_{41})^2 + (e_{12} - w_{42})^2} = 1,208$$

Actualización de pesos

- Función de vecindad nula → solo se actualizan los pesos de la neurona vencedora

$$W_{ij}(t+1) = W_{ij}(t) + \alpha h \left[e_i^{(k)} - W_{ij}(t)\right]$$

$$W_{21}(t+1) = W_{21}(t) + \alpha h \left[e_{11} - W_{21}(t)\right] = -0,52$$

$$W_{22}(t+1) = W_{22}(t) + \alpha h \left[e_{12} - W_{22}(t)\right] = 0,58$$

Escaneado con CamScanner

Ejercicio 01.2 - Guía

$e_1 = (-0.64, 0.52)$

$|d(w_1, e_1)| = \sqrt{(w_{11} - e_{11})^2 + (w_{12} - e_{12})^2} = 0,761$

$|d(w_2, e_1)| = \sqrt{(w_{21} - e_{11})^2 + (w_{22} - e_{12})^2} = 0,939$

$|d(w_3, e_1)| = \sqrt{(w_{31} - e_{11})^2 + (w_{32} - e_{12})^2} = 1,063$

$|d(w_4, e_1)| = \sqrt{(w_{41} - e_{11})^2 + (w_{42} - e_{12})^2} = 0,627$ ← ganadora

$|d(w_5, e_1)| = \sqrt{(w_{51} - e_{11})^2 + (w_{52} - e_{12})^2} = 0,921$

Actualización de pesos (sólo la neurona ganadora)

$w_{41}(t+1) = w_{41}(t) + \alpha h [e_{11} - w_{41}] = -0,37$

$w_{42}(t+1) = w_{42}(t) + \alpha h [e_{12} - w_{42}] = 0,36$

Ejercicio 01.3 - Guía

$e_1 = (-0.64, 0.98, 0.76)$

$|d(e_1, w_1)| = \sqrt{(e_{11} - w_{11})^2 + (e_{12} - w_{12})^2 + (e_{13} - w_{13})^2} = 1,357$ ← GANADORA

$|d(e_1, w_2)| = \sqrt{(e_{11} - w_{21})^2 + (e_{12} - w_{22})^2 + (e_{13} - w_{23})^2} = 1,566$

$|d(e_1, w_3)| = \sqrt{(e_{11} - w_{31})^2 + (e_{12} - w_{32})^2 + (e_{13} - w_{33})^2} = 1,855$

Actualización de pesos (sólo la neurona ganadora)

$w_{11}(t+1) = w_{11}(t) + \alpha h [e_{11} - w_{11}(t)] = -0,07$

$w_{12}(t+1) = w_{12}(t) + \alpha h [e_{12} - w_{12}(t)] = 0,79$

$w_{13}(t+1) = w_{13}(t) + h\alpha [e_{13} - w_{13}(t)] = 0,48$

Escaneado con CamScanner

Ejercicio Parcial (hago varias iteraciones)

$\alpha = 0,7 \quad h = 1$

1° iteración

$|d(P1, W1)| = \sqrt{(0,17 - 0,7)^2 + (0,86 - (-0,1))^2} = 1,096$

$|d(P1, W2)| = \sqrt{(0,17 - (-0,1))^2 + (0,86 - 0,4)^2} = 0,533$ ← GANADORA

$W_{21}(t+1) = W_{21}(t) + \alpha h [P11 - W21] = 0,327$

$W_{22}(t+1) = W_{22}(t) + \alpha h [P12 - W22] = 0,722$

2° iteración

$|d(P2, W1)| = \sqrt{(-0,44 - 0,7)^2 + (-0,88 - (-0,1))^2} = 1,381$

$|d(P2, W2)| = \sqrt{(-0,44 - (-0,1))^2 + (-0,88 - 0,4)^2} = 1,324$ ← GANADORA

$W_{22}(t+1) = W_{22}(t) + \alpha h [P_{22} - W_{22}(t)] = -0,496$

$W_{21}(t+1) = W_{21}(t) + \alpha h [P_{21} - W_{21}(t)] = -0,338$

3° iteración

$|d(P3, W1)| = \sqrt{(0,86 - 0,7)^2 + (-0,42 - (-0,1))^2} = 0,412$ ← GANADORA

$|d(P3, W2)| = \sqrt{(0,86 - (-0,1))^2 + (-0,42 - 0,4)^2} = 1,340$

$W_{11}(t+1) = W_{11}(t) + \alpha h [P31 - W11(t)] = 0,832$

$W_{12}(t+1) = W_{12}(t) + \alpha h [P32 - W_{12}(t)] = -0,324$

4° iteración

$|d(P4, W1)| = \sqrt{(-0,86 - 0,7)^2 + (0,76 - (-0,1))^2} = 1,781$

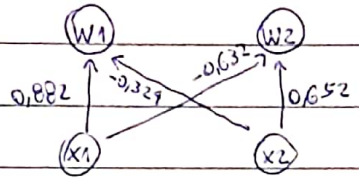
$|d(P4, W2)| = \sqrt{(-0,86 - (-0,1))^2 + (0,76 - 0,4)^2} = 0,840$ ← GANADORA

Escaneado con CamScanner

$w_{21}(t+1) = w_{21}(t) + \alpha h \left[p_{21} - w_{21}(t) \right] = -0,632$

$w_{22}(t+1) = w_{22}(t) + \alpha h \left[p_{22} - w_{22}(t) \right] = 0,652$

Arquitectura de la Red



Ejercicio Parcial

$p1 = (0.17, 0.86)$ $\qquad d(P_i, w) = \sqrt{\sum_{i=1}^{n} (w_{ji} - e_i^{(k)})^2}$

$d(w1, p1) = \sqrt{(w_{11} - p_{11})^2 + (w_{12} - p_{12})^2} = 1,553$

$d(w2, p1) = \sqrt{(w_{21} - p_{11})^2 + (w_{22} - p_{12})^2} = 0,657 \leftarrow$ GANADORA

Actualización de pesos $\qquad w_{ji}(t+1) = w_{ji}(t) + \alpha h \left[e_i^{(k)} - w_{ji}(t) \right]$

* Primero calculamos las distancias a la neurona vencedora

$h(w1, w2) = \dfrac{1}{1 + \sqrt{(w_{21} - w_{11})^2 + (w_{22} - w_{12})^2}} = 0,414 \qquad h(w2, w2) = 1$

$w_{11}(t+1) = w_{11}(t) + \alpha h \left[p_{11} - w_{11}(t) \right] = 0,480$

$w_{12}(t+1) = w_{12}(t) + \alpha h \left[p_{12} - w_{12}(t) \right] = 0,009$

$w_{21}(t+1) = w_{21}(t) + \alpha h \left[p_{11} - w_{21}(t) \right] = 0,17$

$w_{22}(t+1) = w_{22}(t) + \alpha h \left[p_{12} - w_{22}(t) \right] = 0,86$

Escaneado con CamScanner

$$W_1 = [0{,}480,\ 0{,}004] \quad (1,\ 2)$$

$$W_2 = [0{,}17,\ 0{,}86]$$

$$P_2 = (-0.44,\ -0.88) \quad (1,\ 2)$$

$$d(W_1, P_2) = \sqrt{(W_{11} - P_{21})^2 + (W_{12} - P_{22})^2} = 1{,}275 \leftarrow \text{GANADORA}$$

$$d(W_2, P_2) = \sqrt{(W_{21} - P_{21})^2 + (W_{22} - P_{22})^2} = 1{,}893$$

$$h(W_1, W_2) = \frac{1}{1 + \sqrt{(W_{11} - W_{21})^2 + (W_{12} - W_{22})^2}} = 0{,}523 \qquad h(W_1, W_1) = 1$$

Actualización de pesos

$$W_{11}(t+1) = W_{11}(t) + \alpha h \left[P_{21} - W_{11}(t)\right] = -0{,}49$$

$$W_{12}(t+1) = W_{12}(t) + \alpha h \left[P_{22} - W_{12}(t)\right] = -0{,}88$$

$$W_{21}(t+1) = W_{21}(t) + \alpha h \left[P_{21} - W_{21}(t)\right] = -0{,}149$$

$$W_{22}(t+1) = W_{22}(t) + \alpha h \left[P_{22} - W_{22}(t)\right] = -0{,}050$$

$$W_1 = [-0.49,\ -0.88] \quad (1,\ 2) \qquad W_2 = [-0.149,\ -0.050] \quad (1,\ 2)$$

$$P_3 = (0.96,\ -0.42) \quad (1,\ 2)$$

$$d(W_1, P_3) = \sqrt{(W_{11} - P_{31})^2 + (W_{12} - P_{32})^2} = 1{,}473$$

$$d(W_2, P_3) = \sqrt{(W_{21} - P_{31})^2 + (W_{22} - P_{32})^2} = 1{,}169 \leftarrow \text{GANADORA}$$

$$h(W_2, W_2) = 1 \qquad h(W_1, W_2) = \frac{1}{1 + \sqrt{(W_{11} - W_{21})^2 + (W_{12} - W_{22})^2}} = 0{,}532$$

Escaneado con CamScanner

Actualización de pesos

$$w_{11}(t+1) = w_{11}(t) + \alpha h \left[p_{31} - w_{11}(t) \right] = 0,3048$$

$$w_{12}(t+1) = w_{12}(t) + \alpha h \left[p_{32} - w_{12}(t) \right] = -0,635$$

$$w_{21}(t+1) = w_{21}(t) + \alpha h \left[p_{31} - w_{21}(t) \right] = 0,96$$

$$w_{22}(t+1) = w_{22}(t) + \alpha h \left[p_{32} - w_{22}(t) \right] = -0,42$$

Escaneado con CamScanner